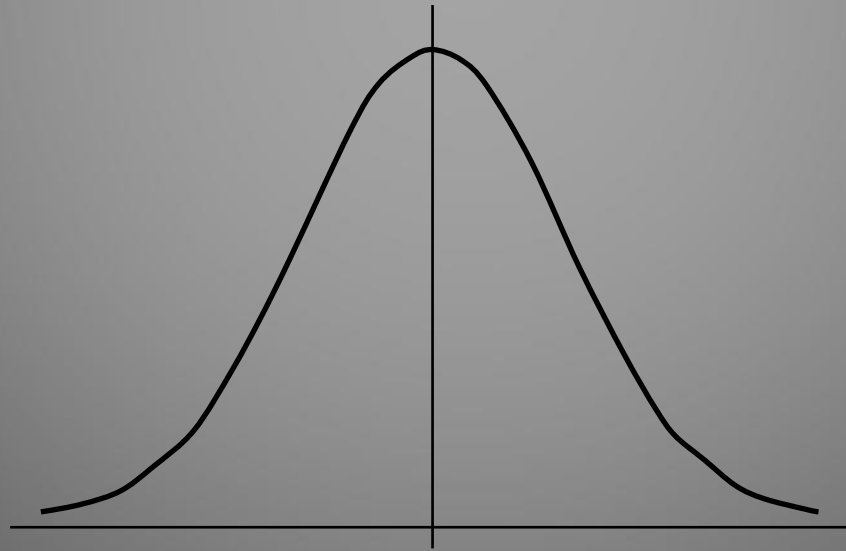


The Normal Distribution

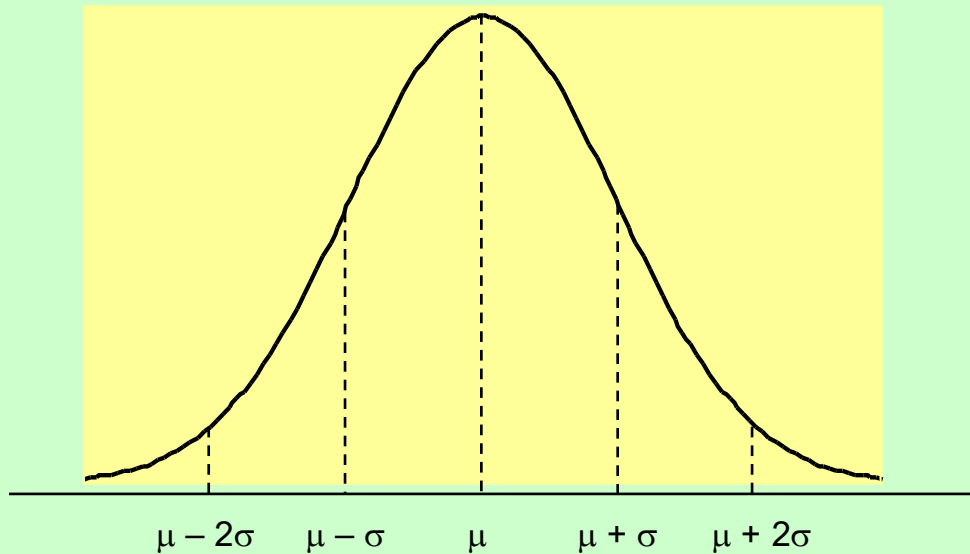


The Normal Distribution

- Very important distribution in statistical theory
- Many phenomena around us and in nature are modeled by a normal distribution
 - Quality characteristics of finished products (e.g., net content)
 - Heights of males/females
- “Convenient” mathematically

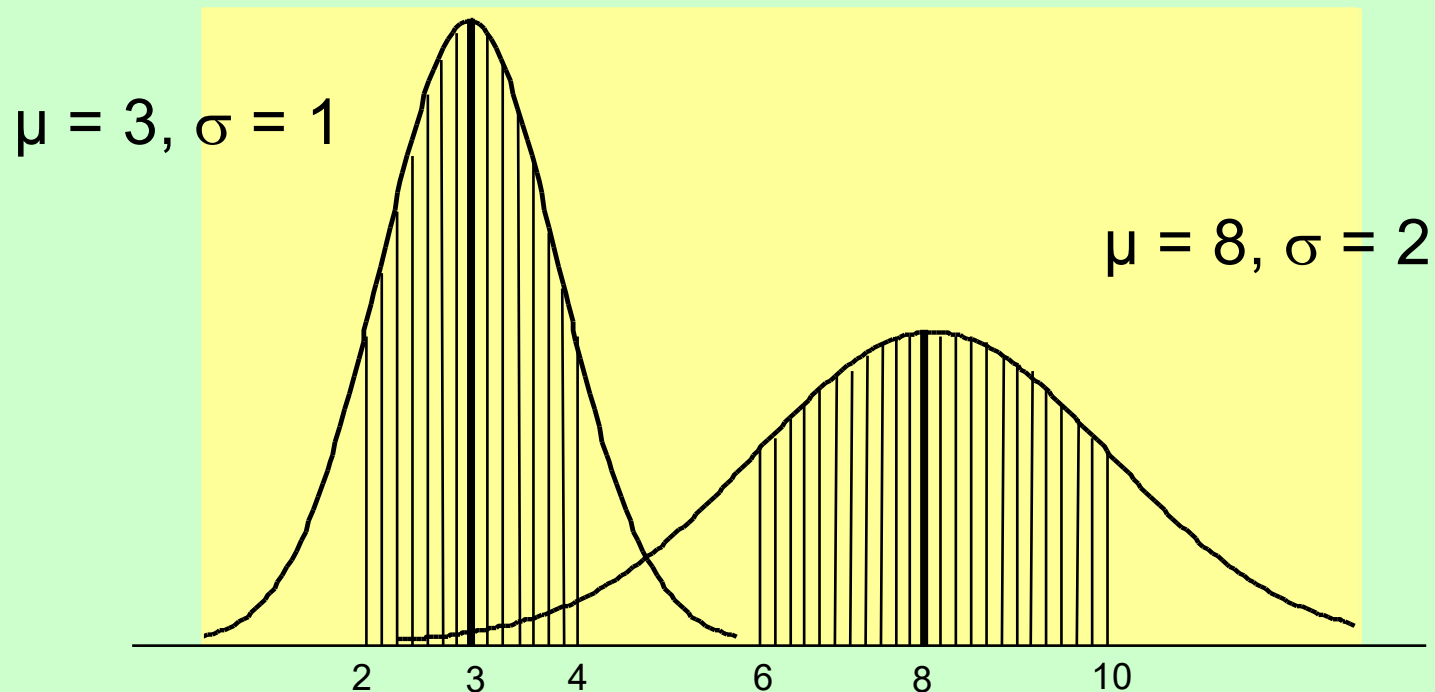
The Normal Distribution

- Is a bell-shaped, continuous distribution
- Has two parameters:
 - μ = mean
 - σ^2 = variance (often the SD σ is given)



The Normal Distribution

- The only thing that matters: how many standard deviations away from the mean
 - Hatched area is the same under both curves



Computing Probabilities

- Goal: to find $P(a \leq X \leq b)$, $P(X \leq b)$, etc.
 - (don't care about " $\leq$ " or " $<$ ") – why not?

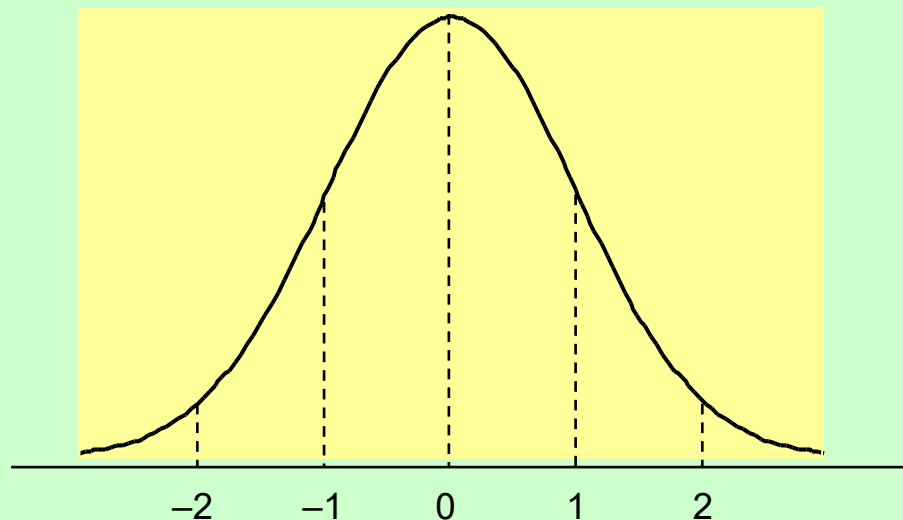
- We know that

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$$

- To find $P(X \leq b)$, can use
 - Minitab
 - Excel

The Standard Normal Distribution

- Denoted by Z
 - Mean = $\mu = 0$
 - Standard deviation = $\sigma = 1$



- Can convert any question about *general* normal RV to a question about *standard* normal RV

Z

- When X is a normal RV with parameters μ and σ , the z-score of X is

$$Z = \frac{X - \mu}{\sigma}$$

- Resulting Z is a standard normal RV
 - $E(Z) = 0$
 - $\text{Var}(Z) = 1$
- Tells us how many standard deviations away from the mean we are (*standardizing*)

The Standard Normal Distribution

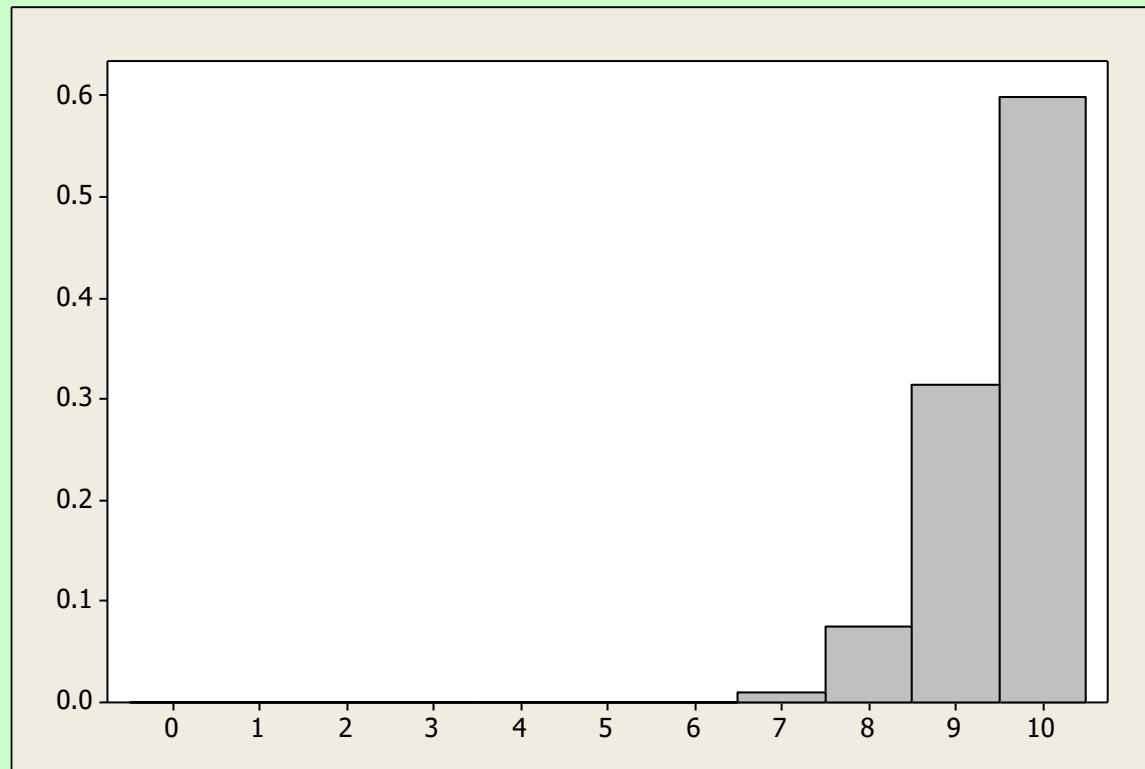
- To compute $P(0 \leq Z \leq z)$ can use
 - Minitab
 - A table
- Examples:
 1. $P(0 \leq Z \leq 1.35) = ?$
 2. $P(Z > 1.46) = ?$
 3. $P(Z \leq -0.82) = ?$
 4. $P(-1 \leq Z \leq 1.14) = ?$
 5. Find k such that $P(0 \leq Z \leq k) = 0.4$
 6. Find m such that $P(-m \leq Z \leq m) = 0.6$

Computing Probabilities – Example

- The daily demand for coke cans in a cafeteria is approximately normally distributed with mean 150 and standard deviation 20. Let X be a RV that represents the demand.
 1. $P(X \leq 200) = ?$
 2. $P(X \geq 180) = ?$
 3. $P(150 \leq X \leq 180) = ?$
 4. How many cans should be stocked so that $P(\text{running out of cans}) = 0.1$?

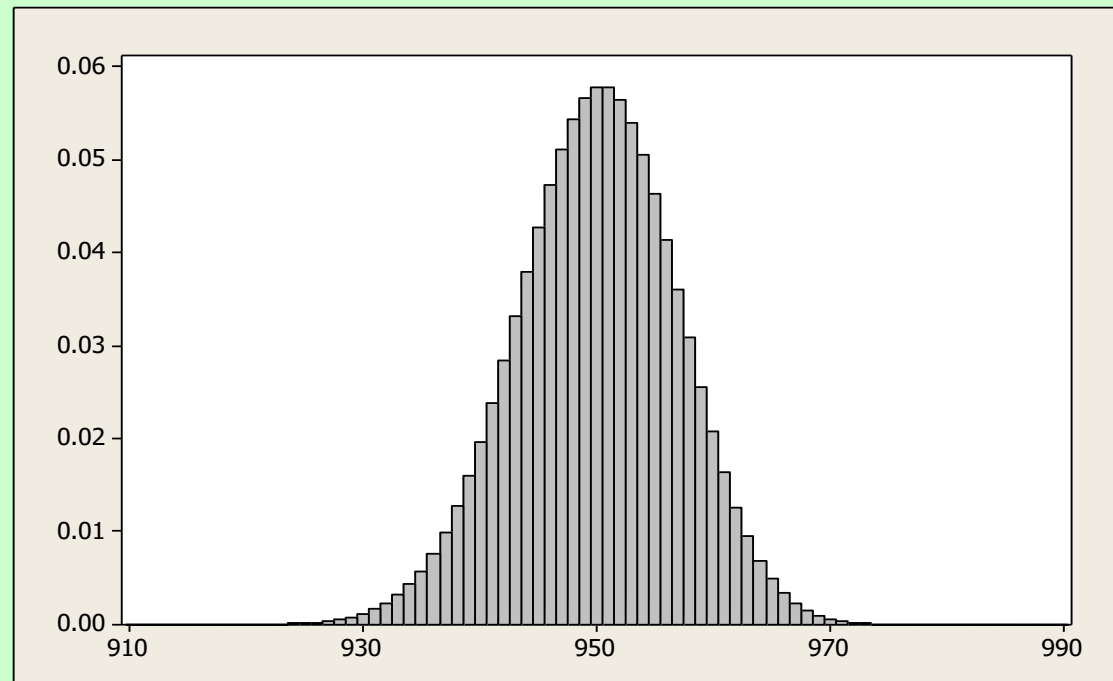
The Central Limit Theorem

- The number of non-defective TVs in a daily sample of 10 has the following distribution:



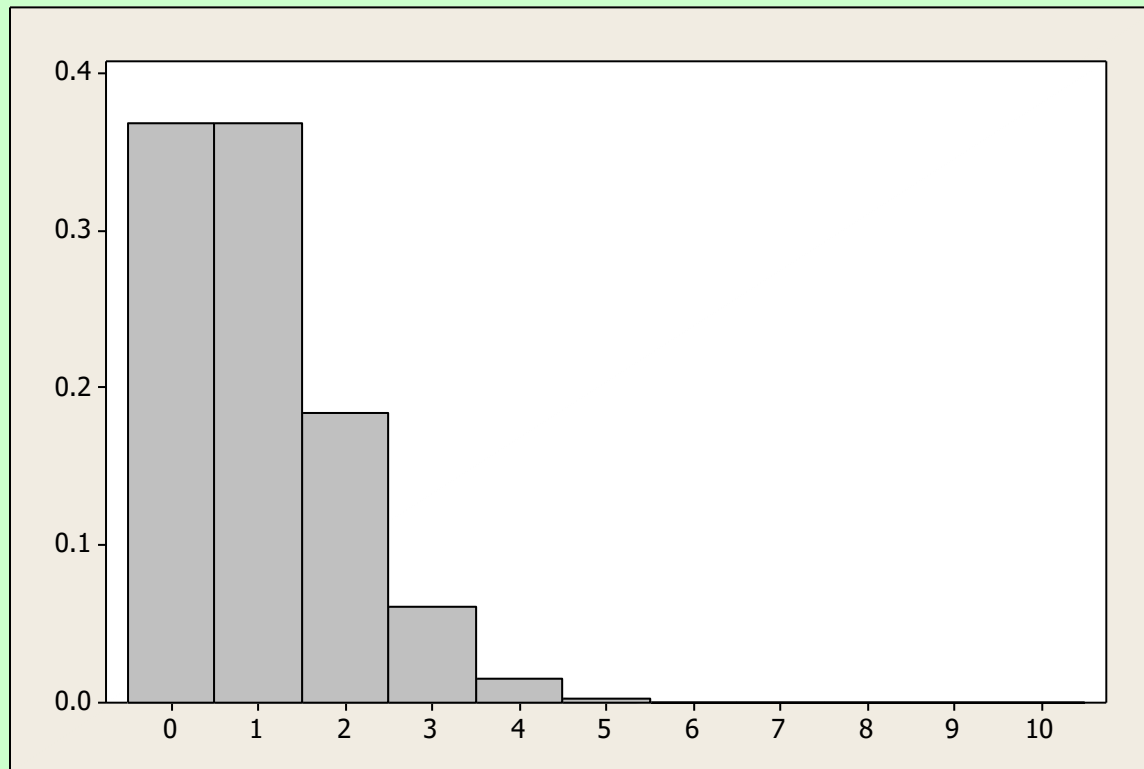
The Central Limit Theorem

- Q: What is the total number of non-defective TVs sampled in 100 days?
- A: It's the sum of 100 independent RVs as before:



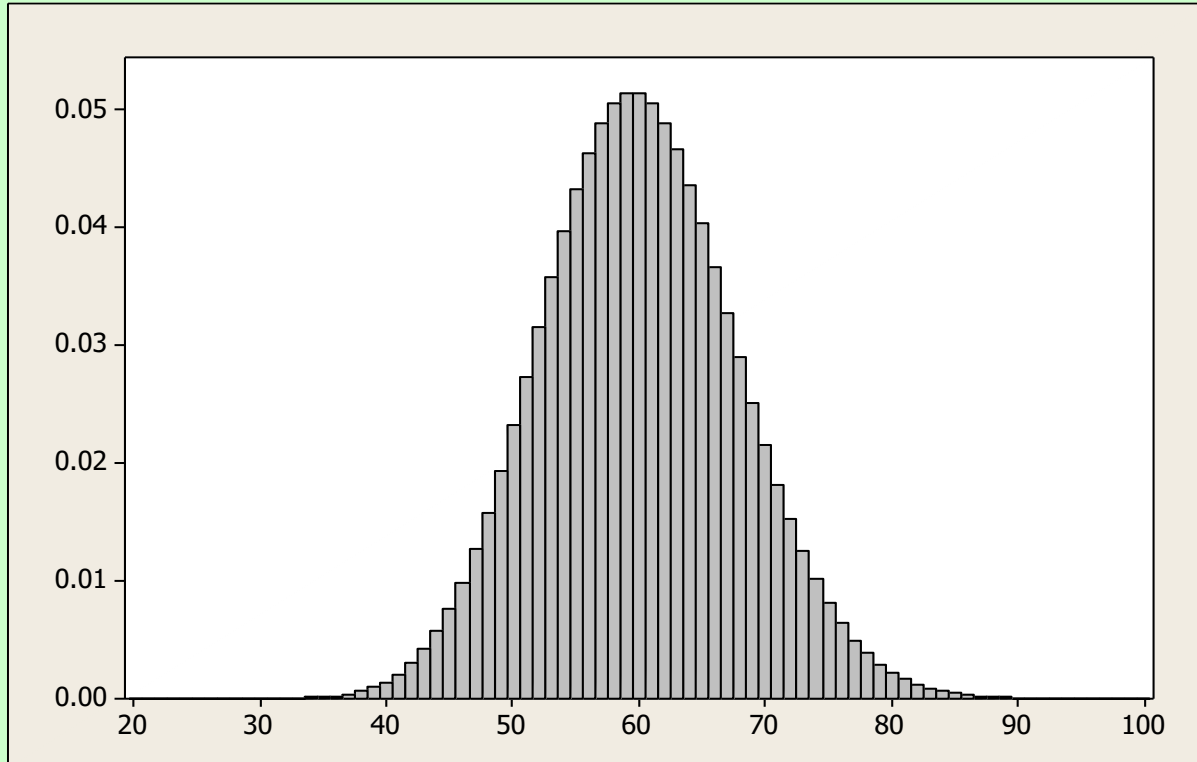
The Central Limit Theorem

- The daily number of homicides in a large city has the following distribution:



The Central Limit Theorem

- Q: How many homicides are there in 2 months?
- A: It's the sum of 60 independent RVs as before:



The Central Limit Theorem

- Q: What is going on here?
- A: The *Central Limit Theorem* (CLT) explains this phenomenon:

The sum of many independent RVs, all having the same distribution, is approximately normal!

- Arguably the most important result in probability theory
- The point: the underlying RVs are not normal, yet their sum is

The Central Limit Theorem

- More precisely: suppose that $X_1, X_2, \dots, X_n$ are RVs
 1. all having the same distribution, with mean μ and variance σ^2 ; and
 2. they are all independent; and
 3. n is large
- Then $X_1 + X_2 + \dots + X_n$ is approximately normal with mean $n\mu$ and variance $n\sigma^2$
 - The SD of the sum is $\sqrt{n}\sigma$
 - The surprising thing is that the sum is *normal*

The Central Limit Theorem – Example

- The daily number of cars sold in a certain dealership has the following distribution:

x	0	1	2	3
P(X = x)	0.1	0.3	0.4	0.2

- Find $E(X)$ and $\text{Var}(X)$
- Assume 300 **days** in a year. $P(\text{Fewer than 495 cars sold in a year}) = ?$

How Large is Large Enough?

- Generally, $n \geq 30$ is good (but there are exceptions)
- When the underlying RVs (the X_i 's) are normal, *any* n is good, and the result is exact (not an approximation)
 - This is not the CLT, but it's true
 - Example: the net content of a soft drink bottle is normally distributed with mean 22 oz and SD 0.15 oz. $P(\text{combined content of 2 bottles} > 43.8) = ?$

Normal Approximation to Binomial RVs

- When X is binomial, with

- $n\pi > 5$ and
- $n(1 - \pi) > 5$,

we can approximate X with a *normal* RV having mean $n\pi$ and variance $n\pi(1 - \pi)$

- Example: The probability that a claim filed with a certain insurance company is fraudulent is 0.06. What is the probability that out of the next 1000 claims to be filed, fewer than 70 will be fraudulent?

The Central Limit Theorem for Averages

- Suppose that $X_1, X_2, \dots, X_n$ are RVs
 1. all having the same distribution, with mean μ and variance σ^2 ; and
 2. they are all independent; and
 3. n is large
- Then $(X_1 + X_2 + \dots + X_n)/n$ is approximately normal with mean μ and variance σ^2/n
 - The SD of $(X_1 + X_2 + \dots + X_n)/n$ is $\sigma/\sqrt{n}$

The CLT for Averages – Example

- The number of employees at branches of a supermarket chain has mean 50 and standard deviation 15.
 1. What is the probability that the average number of employees in Manhattan's 40 branches will be above 55?
 2. What is the probability that the average number of employees in NY's 100 branches will be above 55?
 3. What did we have to assume to answer the above?

Avoiding the CLT Pitfalls

- Be careful not to confuse variances and standard deviations
- Distinguish between *sums* and *averages*, and use the appropriate version of CLT
- When to use?
 - When dealing with a sum/average of a *large* number of RVs
 - When dealing with binomial RVs with appropriate parameters